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To cite this article: Dorsa Kafili & AnnMarie Brennan (2024) Architectonics of Videogame Sound Design: The Use of Sound and Music in Creating Immersive Virtual Spaces, *Architecture and Culture*, 12:3-4, 439-466, DOI: [10.1080/20507828.2025.2530892](https://doi.org/10.1080/20507828.2025.2530892)

To link to this article: <https://doi.org/10.1080/20507828.2025.2530892>



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Published online: 15 Aug 2025.



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ARCHITECTURE AND CULTURE

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Keywords: architectural
representation, sound and
space, soundscape,
videogame sound design,
videogame music, immersion



Volume 12/Issue 3-4
September–December 2024
pp. 439–466
DOI:10.1080/20507828.2025.
2530892

No potential conflict of
interest was reported by
the author.

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Architectonics of Videogame Sound Design: The Use of Sound and Music in Creating Immersive Virtual Spaces

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ABSTRACT Sound is both an atmospheric and functional element of understanding one's orientation within a space, whether real or virtual, evoking a sense of spatial awareness. With videogame engines becoming increasingly prevalent in architectural practice for constructing virtual spaces, coupled with the growing interest of architects in videogame design, an opportunity emerges for the intentional designing of sound as an essential spatial component. By examining the role of audio in videogame environmental storytelling, this article provides insights for environmental designers, world-building artists, and architects seeking to comprehend dynamic sounds to narrate their digital stories using interactive and non-linear storytelling media. It does so by illustrating how videogame sound designers and music composers create and integrate sonic experiences that respond to in-game environments in three case studies. This research was conducted through a series of interviews with leading videogame sound

designers and music composers, regarding the details behind the auditory components of videogame world-building.

Introduction: The Sound of Spaces

Audio-visual modes of spatial representation in many contemporary videogames often feature spatial and architectural designs as part of their virtual world-building to support immersion. As defined in videogame and VR (virtual reality) scholarship, immersion refers to the degree to which a virtual simulation engages multiple sensory modalities and triggers human senses.¹ Immersion is “the objective level of sensory fidelity” delivered by a system—such as a virtual reality platform or interactive simulation in videogame media. It is distinct from presence, denoted as the “subjective psychological response of a user” experiencing the same system.² For the context of this article, the notion of immersion refers to the intentional shaping of sensory qualities designed to maximize engagement of an individual within that experience, where audio is “the catalyst [for] self” in the immersive systems of videogame design.³

In videogames, immersion is accomplished by utilizing different visual attributes such as architectural form, spatial arrangement, virtual tectonics, and light schemes, which constitute a shared language between the two disciplines of architecture and videogame design. While sound is omnipresent in architectural space, game designers are required to incorporate designed sound and music to create engaging virtual spatial experiences. This article aims to establish tangible connections between sound, music, and the creation of virtual spatial environments. It explores sound and music as significant parameters that work with the virtual setting to shape the videogame imagery, spatial understanding, and orientation within the game world.

This research analyzes and details methods of sound and music creation in videogames by employing a designer-focused perspective to elaborate on how videogames manifest higher degrees of immersion through sound design and music composition in relation to context. This exploration was conducted through interviews with game sound designers and music composers of three games: James Therrien from *Rain World* (Videocult, 2017), Yann Van der Cruyssen from *Stray* (Blue Twelve Studio, 2022), and Austin Wintory from *Journey* (Thatgamecompany, 2012). Experiencing the virtual world through non-human characters, these games derive significance from a minimal use of dialogue, which enhances the palpable links between sonic experiences and the virtual space, and prioritizes sound, music, and spatial design as key vehicles in driving the game narrative.

As the defined professional boundaries regarding the scope of work for architects become more and more expansive and ambiguous—where designing for virtual, metaverse realms is increasingly significant as real-world scenarios—architects share the same creative and

technical ground as world-building designers and environmental artists.⁴ Therefore, they are required to gain insight into the designing of sound for these virtual, immersive environments. Moreover, there is a growing interest among architects in designing videogames or game-like spaces.⁵ Given this escalating trend, Marc Bonner, a scholar specializing in videogames and media, contends that architecture “has to be understood anew in its medial possibilities and how it can mediate and regulate the environment to the observer by enabling new spheres of action.”⁶ With architecture becoming open to adopting videogame design elements to define virtual spatial structures, it is crucial to comprehend specific parameters inherent in these fields and effectively integrate them into the realm of architecture. Therefore, in order to effectively adopt this medium into the design processes of architectural spaces, it is essential not only to comprehend videogame spaces through the lens of architectural design methods but also to unveil their distinct parameters from the perspective of a game designer.

By presenting insights gained from interviews and analyzing responses related to their practice in developing games, this article facilitates the transfer of technical knowledge from the field of videogame design to architecture, while establishing a shared vocabulary that bridges the design processes between the realms of these two disciplines. Following multidisciplinary scholars in architecture and videogames, who suggest that the inherent spatiality in videogames establishes the medium’s value as a tool for architectural design, this article opens up opportunities for architects and environment artists to integrate dynamic sound and music as critical elements in interactive representations.⁷ Since the medium of videogames offers new possibilities for representing engaging digital spaces, this article promotes the discussion of sound, a spatial characteristic that architects and environmental artists should invest more in as a design tool to create impactful depictions of their conceptions.

Background: “Aural Architecture”

Sound and music are significant atmospheric, experiential, and expressive spatial elements in both physical and virtual realms. The pioneering composer and acoustic ecologist R. Murray Schafer formulated the overarching notion of soundscape to underscore how an environment’s acoustic structure shapes the overall spatial atmosphere. Sound is influenced through its interaction with different forms, objects, surfaces, textures, and materials, which in turn affect perception. In his view, sound and soundscape are not just passive causations of space but actively engage with and transform it. Introducing the idea of “soundscape design,” Schafer advocated the intentional modification of the acoustics of an environment to enhance the experience within it.⁸

Schafer noted that modern design became utilitarian to the extent that he deemed “architecture [had] ceased to be the art of positive

acoustic design.”⁹ Barry Blesser and Linda-Ruth Salter, authors of *Spaces Speak, Are You Listening?* (2007), agreed with Schafer’s sentiment and revived the notion of acoustic awareness in architectural spaces. They postulated “aural architecture,” suggesting that sound itself can be an intentional design component that contributes to the shaping of a space. Blesser and Salter argue that the esthetics and symbolism of aural architecture “parallels visual architecture,” evoking specific experiences through auditory cues and sonic attributes.¹⁰

The acoustic engineer Trevor Cox discusses the value of a “soundwalk;” a literal walk with the sole intention of carefully listening to connect with the sensations of hearing. This allows discovery of unknown auditory treasures “to experience the most surprising, unexpected, and sublime sounds” that excite and inspire creativity.¹¹ Similarly, Michael Fowler, a scholar in the field of soundscape studies, formalizes this notion as “critical listening,” an approach to reading sounds and simulating acoustic spaces “through the development of an informed and constructive auditory imagination.”¹² They encourage architects to utilize other senses that might offer more diverse ways of experiencing built environments.

These links between sound and spatial design are replicated in music. While there is no argument that architectural space affects the acoustics of how music is heard, David Byrne, artist and music theorist, has explored how the reciprocal relationship between music and architecture has historically shaped the evolution of musical forms. He suggests that certain types of music emerge because of the acoustics generated by specific spaces.¹³ These scholars all assert that sound and music influence our cognitive perception of spaces, making auditory considerations a crucial component of design in architectural form.

Sound and music in immersive spaces of videogames are often crafted with even greater precision than physical architectural projects, with game designers exercising more control over the auditory experience. “Space is no longer just a geographic framework (near-far, front-back, up-down) for positioning sounds relative to listeners,” nor simply a “response to the acoustics of the environment.”¹⁴ Instead, “aural architects” of virtual spaces actively manipulate listeners’ cognitive maps and the feel of a virtual realm.¹⁵ These reciprocities and varied approaches to listening experiences in physical environments can thus be expanded to virtual representations of spaces where designers decide how sound and music are imagined and experienced.

Case Studies: An Alternative Sense of Space through Sound and Music

The opening scene in the indie videogame, *Rain World*, uses momentary sequences combined with a soundtrack to narrate the story of a chimera called Slugcat who was separated from its family and lost amongst the remnants of a ruined industrial environment composed of broken machinery and decaying structures. *Stray*, another indie-developed

videogame, yet different in representational style, utilizes the same nonverbal audio-visual strategies to narrate the story of a feline traversing a dense cyberpunk urban setting. Using synthetic sounds and found object music, the sonic experiences adhere to the post-apocalyptic and dystopian worlds they represent. In contrast to these two games, *Journey* employs more traditional musical instruments and orchestral scores to narrate its landscape. Together with the visual aspects, the game places the player in a vast desert. With no dialogue or script, *Rain World*, *Stray*, and *Journey* must narrate their opening storylines through musical and acoustic means that align with the spatial context. The sound design and the musical approach in these games integrate with the worldbuilding representations to give players a sense of narrative depth that can be expected in a digital work or a game.¹⁶ As such, by employing these two dominant perceptive modalities, visuals and audio, namely videogame designers seek to reinforce a sense of immersion.

In *Video Game Spaces* (2008), media scholar Michel Nitsche categorizes game audio into four components: sound effects, soundspace, music, and speech. Similarly, Karen Collins, author of multiple books on videogame audio, uses a comparable framework. However, in her discussions, the term sound encompasses the broader range of sonic elements, including “discrete sound effects, ambient sound beds, dialog, music, and interface sounds.”¹⁷ This article adopts the same general categorization of game sound. Nevertheless, as the case studies examined here use minimum dialogue with a focus on relations between sound, music, and environment, this article contributes specifically to environmental storytelling, rather than narrative-driven storytelling through dialogue.

The exploration of each case study begins with an overview of the game before elaborating on the audio techniques in relation to the game space. The majority of the following quotes come from direct interviews with Therrien, Van der Cruyssen, and Wintory. This investigation includes gameplay footage and a combination of story and soundboards to further elucidate these concepts.

In analyzing each case study, presented soundboards feature a series of visuals, explanations, and sound graphs of the corresponding game scene. Each key moment is broken down into short, captioned segments in a comic book-inspired format, designed to illustrate how sound and music evolve in real-time in response to the player’s progression throughout a section of the game. The visual and textual narrative captures the dynamic interplay between gameplay and audio while emphasizing the emotional and spatial shifts that occur moment to moment.

Rain world: “alphas, gems, and junk” (footage: https://youtu.be/M_cXbweNajw)

Rain World is the rebirth of a new world sprouting amongst the debris of an older fallen one, and deals with an alien world of unknown creatures.

It is based on the idea of “rummaging through the remains of a failed society” and experiencing its crumbled structures.¹⁸ Represented in two-dimensions, the environment of this adventure, *Metroidvania*, depicts the “aesthetics of decay.”¹⁹ While initially aiming to create a procedurally generated world, designers of *Rain World* expanded their focus to encompass atmospheric characteristics of a virtual space through “a bespoke approach where it required environmental storytelling.”²⁰ While the objective of the game is for the main character to find its way back to its family, the game features a unique experience of atmospheric space and moods along a journey that comprises rain-world. The esthetic goals of the game are accomplished through the creation of images, sounds, and music, which were later compiled into an album titled, *Alphas, Gems, and Junk*.

Architectural theorist Juhani Pallasmaa highlights the relationship between image, imagination, and multi-sensory experiences of spaces. He asserts the prominence of sound compared to vision in stimulating imagination: “Sight isolates, whereas sound incorporates; vision is directional, whereas sound is omnidirectional. The sense of sight implies exteriority, but sound creates an experience of interiority.”²¹ How auditory icons, including the sounds of characters, objects, environment, and interactables operate in a game to convey meaning is determined by the “modes of listening.”²² Reduced, casual, semantic, imaginative, and navigational listening experiences are different ways of hearing in virtual spaces which can vary depending on context and user experience.²³ In *Rain World*, the shifts and flows of visual and aural representations affect the pace of navigation and actions of the player, leading to conjuring the modes of listening in its space and conveying different relationships of the surrogate body and virtual environment.

When designing fantastic, imaginative, futuristic, or sci-fi settings, the spatial experience must feel possible and relatable, with sound and music being essential in achieving this. In representations of imaginary worlds, sound, soundscape, and music contribute to immersive experiences by “establishing a concrete and believable reality.”²⁴ Audio design within a virtual environment should consistently fit or correspond to the context. Crafting a reality together with the modes of listening can later be contrasted by introducing sounds and music that would not naturally emerge from the represented environment, such as the “threat music” when encountering an enemy or horror sounds. A sense of immersion involves players accepting the in-game representations as credible stimuli, without fighting “to suspend this natural tendency to ‘believe’ and determine the referential level of the objects they are perceiving.”²⁵ Fostering immersion through sound design and music that consistently corresponds to the context enhances the acceptance of fantastical audio elements within a virtual world.

The “imaginative immersion” mode of listening in game spaces contributes to contextualizing the player’s character within the game.²⁶ In

Rain World, imaginative immersion helps the character to act upon the scenery depicted in the level design, inviting the rhythm and pace of players' actions in certain locations. This immersive dimension works closely with visuals to evoke the notion of ambiguity, offering players moments to capture the essence of representation through auditory experiences. Being responsible for both sound and level design, James Therrien had unrestricted artistic freedom, allowing him to craft sound and music for the context, and sometimes the context to complement these auditory elements.²⁷ In the interview, Therrien explicitly highlights this reciprocal creation:

We would use a ladder for climbing, pairing it with a musical composition that creates a sense of wonder and expectation. Then, maybe, atop the climb, I would have a scenic vista where the player would get a moment to take a breath and consider what they were seeing. Sometimes, I would have a piece of music in mind that I really liked, that I wanted to have a moment with, and I would be able to design a level or a lead up to where this music would play to heighten the experience or lower the expectations for the player to then give them some sort of release. So, for me, it was a lot of using the tools that I use for music, which is building expectations, use of repetition and variation and kind of spacing things out in a way that gives players moments of space to reminisce or contemplate. I think that sort of rhythm really informs the overall experience because the pacing of the game is very specific for me; it has definite builds and peaks and valleys of both excitement and exploration, and I was able to orchestrate around that, which was an extremely unique and fun experience.²⁸

Coming from a background in musical studies, with years of experience in performing chiptune music, Therrien began experimenting with various “junk” sounds sourced from literal junk materials to create diegetic and nondiegetic sound and music of the game.²⁹ Introduced from cinema to videogame sound theory, these two categories of audio refer to aural experiments in representations of space that either emanate from the on-screen fictional world and are heard by the character within the game space (diegetic) or added externally to enhance the viewer's experience that is only heard by the audience or the player (nondiegetic or extradiegetic).³⁰ In videogames, player's interactivity and dynamic audio challenge the conventional boundary between diegetic and non-diegetic sounds found in cinema, as player-driven actions blur the line between sounds that belong to the game world and those that exist outside of it.³¹

The main idea for diegetic and nondiegetic musical languages in *Rain World* is “re-evolved out of a fallen world of old rusted vehicles and

worn-down computers” to represent different ranges of soundscapes from familiar and simplistic ambient sounds to eerie and otherworldly.³² Inspired by the organization of sound and structurally “aleatory” aspect of John Cage’s music creation, a lot of the “found object music” in *Rain World* is built from banging on old rusty pipes, recording squealing computer sounds, and gathering samples of distorted voices to transform them into a chorus.³³

Transitions between diegetic and nondiegetic sounds contribute to forming videogame esthetics. In *Rain World*, the in-game pulsating wind sounds blend seamlessly into a piece of music that only the player hears. This passage plays a pivotal role in shaping the narrative, atmosphere, and emotional impact of the representation of the space on a screen, as well as the gameplay aspects. Referred to as “fantastical gap,” diegetic and nondiegetic audio denote transformation and imagination where reality and fiction coexist, intertwine, and present storytelling opportunities related to visualizations to enrich the depth of the overall experience.³⁴ Audio transitions also play a role in world-building by depicting the ambient sounds that reflect the designed environment or setting.

Rain World begins with a montage of illustrations depicting the story of the protagonist set against subtle synthesized music, thematizing distinct musical sounds within a futuristic timeframe. Soon after entering the game world, the simple ambient sounds introduce and situate the player in the abandoned environment with only a few sound sources. Halfway through the game, the player reaches an enclosed space-tech with massive structures and an incomprehensibly giant supercomputer that stands as testimony to the relics of a long-gone civilization. Namely, the General System Bus is a subarea within the supercomputer known as Five Pebbles. While in harmony with the rest of the game world, the ethereal nature of this tech environment changes significantly in quality and marks a profound departure from the familiar esthetics of the game and gameplay.

Five Pebbles starkly contrasts with the other world biomes. It is reflected in the representations of color, scale, lighting, materials, sound, and dramatic changes in the movement through space by defying gravity. To enter this region, a gate creaks open with a loud, echoing noise, signaling the beginning of the player’s action after a short cutscene image in the safe zone (Figure 1a). The soundscape of the environment, including rises and falls of ambient sounds such as the howling wind in the tunnels, combined with the distinct characteristics of the sound design, perfectly conveys the sense of this semi-closed space. Footsteps and nondiegetic sounds of passing through tunnels dynamically adapt to the player’s actions and input. These sounds also contribute to the consistency of shifting scenes in this platformer game, where scenes change in a fraction of a second (Figure 1b).

Upon entering enclosed spaces in this region, the sound of the rain fades, and prolonged reverberations take the place of the echoes.

Neon lights and their glitching sounds are critical elements, both visually and sonically, to convey the tech nature of this environment (Figure 1c). The presence and absence of gravity are marked by distinctive sounds,

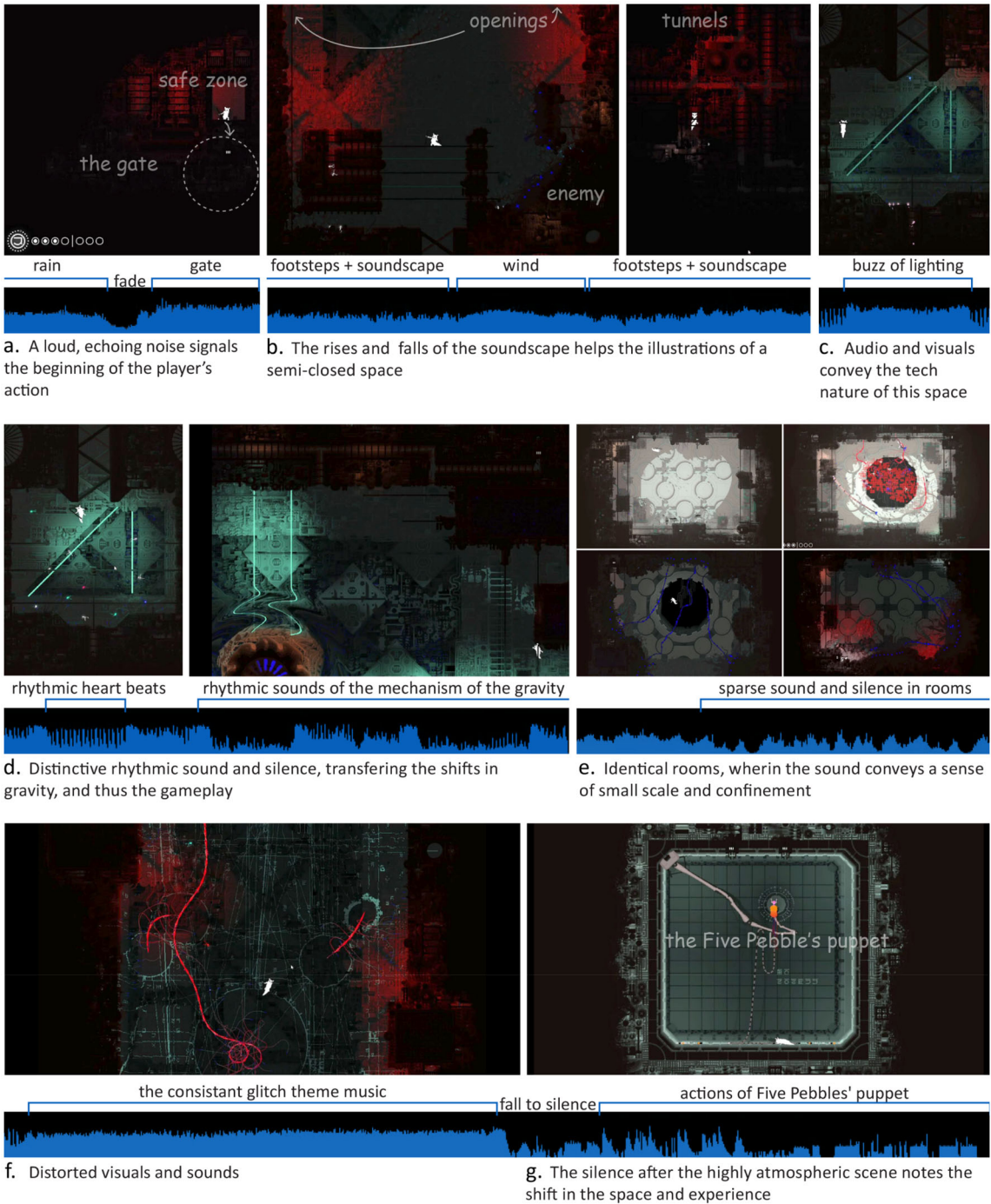


Figure 1
Rain World, visualization of the soundboard, by the author; soundboard footage: <https://youtu.be/DvcCifrKLzM>, courtesy of Videocult.

with continuous pulses resonating throughout a ceaseless symphony of technological waves coursing through space, in harmony with the grid-based design of rooms (Figure 1d). The interval sounds note the functions of the space and follow the actions of the player. The sound of the protagonist's heartbeat reflects the fear induced by the ambiguous nature of this environment and the unpredictable changes in actions that are difficult for the player to control.

As the player delves deeper into the General Systems Bus, the once omnipresent technological cacophony gracefully metamorphoses into harmonious music, transforming their perception of the space and creating a profoundly atmospheric gaming experience. While this cacophony works in harmony with the corroded metal and hissing of the obsolete machine, it is the reverberation caused by the walls that activates immersion in a confined space of the players' minds. Situating the player in identical rooms with certain dimensions marks a dramatic change—a unique and distinctive experience within these rooms. Moments of silence in such areas amplify the prevailing sense of isolation; when rhythmically alternating between sound and silence, this auditory experience infuses a unique high-tech ambience distinct from other environments (Figure 1e).

While in *Rain World* each segmented area divided by camera cuts is identified as a “room,” in *Five Pebbles*, rooms are visually crafted to evoke a genuine sense of confinement through the arrangement of walls, limits, and boundaries. What truly immerses the player and enhances the sense of confinement is the sound design. Although a fictional space that is represented in pixel art style, the embedded acoustic conditions create an embodied experience in a place of a given scale. The sound design intricacies mark transitions across various regions in the level design, signify actions within the world, and sometimes act as the primary indicator of imaginative immersion in the dystopian environment.

Upon the entrance to a room with distorted visuals, the theme music of the game begins to distort, introducing an unexpected twist. Together with the visuals, this scene conveys a profoundly atmospheric experience (Figure 1f). Finally, the silence after the highly atmospheric scene notes the shift in the space and experience. The silence after the highly atmospheric scene notes the shift in the space and experience (Figure 1g).

Turning the “found” sounds into instruments, Therrien manipulated, tweaked, and twisted sounds with synthesizer and computerized processes to create an alien effect. This method was implemented on field-recorded sounds, voice recordings, and real-time audio samples to maintain the sonic unity, underscoring their common origin within one world. Simultaneously, it introduces diversity through various layers of mixing sounds. In the selection of sound sources, Therrien affirms that “we were attempting to capture the aesthetics of the world as we perceived it and illustrate what might naturally emerge from that.”³⁵

In dialogue with Therrien, it is evident that there exists a reciprocal relationship between creating audio and the environment in dual ways. On the one hand, the creation of soundscape in *Rain World* is tied to the incorporation of sounds related to elements present within the scene and sound behaviors in open or closed spaces to impart information about the narrative, settings, and prompting actions. Conversely, there are instances where the design of an environment and its scenery is influenced by the designer's desired tone conveyed through a particular sound and music that he wants "to have a moment with."³⁶ This approach not only defines the space but also influences the rhythm and pace of the gameplay.

The envisioned sense of escapism within the dystopian setting of *Rain World*, narrated through the lens of a non-human perspective, intensifies otherness and introduces ambiguity by playing with expectations and crafting an immersive experience within the "moment of interpretation."³⁷ Therrien asserts that "*Rain World* is very sparse, and the music is not necessarily just from within the game space, but it plays with your expectations of that."³⁸ Establishing moments of interpretation through sound design and in relation to the architecture of the game is extremely important in the ongoing experience of "other" spaces, as they contrast reality and immerse players within environments that cannot possibly be real.

In *Rain World*, sound and music contribute to enriching the sense of immersion in the game's pixel art-represented space, developing an atmospheric experience that accompanies the visuals. Its audio engages an imaginative mode of listening that blends familiarity with eeriness, and at times it subtly dictates the player's behavior and encourages specific actions in response to sonic cues and imagery. In essence, sound creation corresponds to visuals and simultaneously guides imaginary immersion according to the world-building, even in a platformer mode of representation, where the player is detached from the scene.

***Stray: Morusque's music sheets in The Slums* (footage: <https://youtu.be/hHweUY1EgEw>)**

Experimentation with instruments, recording ambient sounds, and synthesizing them occurs in a similar fashion in *Stray*, a game that portrays another animal experience amongst cramped buildings filled with machines and robots. A point of difference between the two games is the progression of the narrative using a form of written dialogue. While the voices of denizens in this game are intentionally designed to sound robotic and distinct from human speech, conversations are facilitated by B-12, a robot that formed a friendship with the protagonist very early in the game.

At the outset of their journey, in *The Slums*, the cat and the robot meet Morusque, a robot with a guitar that resembles the sound designer of the game, Yann Van der Cruyssen. This musician robot asks the cat to

find him music sheets to play. The diegetic music in this game mimics traditional human practices.³⁹ While it may not precisely replicate a human playing, there are moments where a jazzy underscore or an imitation of improvisational humanistic music can be discerned. The same happens with synthesized sounds to resemble human voices that have been robotized over time. Represented in a three-dimensional realistic style in the third-person viewpoint, the sound design in *Stray* triggers machinery sounds in a decrepit cityscape, in balance with the cyberpunk city that the game portrays, which references the Kowloon Walled City in Hong Kong. The game employs architecture to create its own reality by “engaging a rigorous process of spatial planning through narrative and ludic techniques.”⁴⁰ While *Rain World*’s emotive storytelling is hinged on audio as a primary concern to portray an entirely alienated world, *Stray*’s sound design contributes to an experience reminiscent of a humanistic society.

At the heart of the sound design of *Stray* was the importance of producing glitch sounds in harmony with the decaying technological world. Using computer and hardware devices like synthesizers, Van der Cruyssen constructed rough and “not clean” audio. He explains, “I owned a synthesizer that was dying at the time, generating strange sounds it was never supposed to do. I also had a device known as a ghost detector, which is a kind of coil, capturing electromagnetic fields and producing glitch audio recordings.”⁴¹ Developing a rich library and processing the recordings, he created auditory icons through mechanical beats, rhythm, and musical scales which contribute to the “sound functions” in virtual spaces.

“Sound functions” express interactivity, space, and time affordances. The perception of diverse spaces through sound, “which is the acoustic volume enveloping and morphing around the player,” has a crucial role in influencing the overall sense of immersion in the game world.⁴² Sound functions offer the affordance to perceive localization through acoustic rules (choroplasts), the directional situation (topoplasts), passing timeframe (chronoplasts), or the chronological context linking to the past, present, or future (aionoplasts).⁴³ In *Stray*, sound functions align with visualizations to convey dystopian, futuristic, and post-apocalyptic time and space.

To note localization, digitally crafted tracks were assembled in a chaotic manner, much like scrambled compositions, making it difficult to discern any distinct melody. In addition, by incorporating reverberation, Van der Cruyssen attempted to convey a sense of space and uniformity across various in-game environments. This involved not only applying reverb to ambient sounds but also extending it to nondiegetic music external to the game world. Reverb control was utilized to ensure the music resonated consistently, creating the impression that it belonged to the same recording space while at the same time providing the feel of scale in a given space. As an esthetic element, he tuned and tweaked the

reverb to impart distinct characteristics to the sounds. At times, he pushed it to extreme settings, transcending traditional reverb, and instead introducing the intended distorted sounds.

The tunnel is the first environment after the player character falls, as an in-between space that catalyzes the post-apocalyptic world (Figure 2a). This area instills an oppressive sense of claustrophobia with its cave-like echoes and isolation by amplifying a small trickle of dripping water and the player’s footsteps. A faint, metallic resonance lingers on the edge of hearing, unsettling the ear and alluding to the unnatural environment. Upon entering the Dead City, one loud sound produces a sophisticated “ding,” signaling the entrance to an unknown environment (Figure 2b). Meandering in the Dead City, the same ambience is enhanced with the inclusion of pulsating, whirring hums and static, punctuated by a powerful “clunk” of something hidden locking into place, as if in response to the player’s presence (Figure 2c). This brief moment of tension falls back to the metallic resonating, now including heavily distorted steel reverberation, reflecting the probable and likely hostile life of the city in



Figure 2
Stroy, visualization of the soundboard, by the author; soundboard footage: <https://youtu.be/djlq8M1C6GE>, credit: BlueTwelve Studio.

hiding (Figure 2d). During the chase sequences, industrial music plays with a driving rhythm at an increased tempo, heightening excitement and tension from the presence of danger. Hints of deep pulses that sound “wet” suggest something organic in amongst all the mechanical music, which is reflected in the fungal, cancerous material growing throughout the shells of abandoned homes (Figure 2e).

Studying general design, music composition, and videogame design, Van der Cruyssen began experimenting with graphic design and programming games. This experience proved valuable when he was later hired to design sound and music while handling the programming aspect of games in 2008, specifically focusing on integration, utilizing custom engines alongside prevalent game engines. For *Stray*, he used the built-in Unreal sound engine, requiring Van der Cruyssen to code and “hack a lot of stuff” to manually place zones of audio or manage complex reverberations to ensure a seamless transition in sound quality when the character moves between indoor and outdoor environments.⁴⁴

Investigating videogame spaces, media scholar Michael Nitsche argues that “as with the virtual camera, there is no ‘natural’ listening position in a video game space.”⁴⁵ Aural perception is omnidirectional and all-encompassing, making it difficult to determine a spatial point of audition. In virtual environments, however, it can be precisely positioned during the design phase. The curation of sound in two-dimensional, axonometric, and three-dimensional spaces requires defining the positioning of the audio and determining when and how it should be played. Videogame sound designers integrate every crafted audible element in the virtual space in a way that enhances the player’s spatial awareness. Van der Cruyssen underscores that in *Stray* the point-of-audition origin for each sound is camera-centered, whereas music triggers are based on the actions and movement of the protagonist. In certain contexts, as the camera begins to rotate around the cat, the viewer hears indoor sounds when the camera is inside, and vice versa, discerning the outdoor soundscape when the camera is situated outside (Figure 3a).⁴⁶

Similarly, diegetic audio streams are influenced by the actions and movements of the cat (Figure 3b). The music emanating from nested devices like the numerous radios distributed within the scenes, for example, fades in and out depending on the cat’s proximity, encouraging players to thoroughly explore new areas. While the music is hinged on the cat’s and the camera’s movements, this dynamic audio enhances spatial awareness and captures a retro esthetic within a decaying cyberpunk city. These diegetic audio loops provide a sense of the mundane setting and add a texture to the game’s narrative by hinting at a world beyond the player’s practical focus.

The same listening position is not applicable in games represented in two dimensions where the sound pressure is heard omnidirectional regardless of its source. In this regard, Michel Chion

notes point-of-audition as the dual concept of perceiving sound within a depicted space, and its connection to an on-screen character experiencing similar spatial and subjective auditory sensations, “a spatial sense” and “a subjective sense” respectively.⁴⁷ Therefore, comprehending the correlations between the point-of-audition, point-of-view, and background levels is crucial when implementing sounds and music using game engines in spatial representations.

Sound integration in game spaces regarding the camerawork leads to diverse listening experiences. Compared to platformer modes of representation, the integration of sound slots in three-dimensional virtual spaces is essential in the reinforcement of embodiment. In the side-view videogame *Rain World*, the integration of audio is designed to align with the static camerawork which restricts the player from zooming in or out to different angles of the screen. Accordingly, the point-of-audition is abstractly positioned at a distance from the image, coinciding with the center of the screen where the perception of sound feels omnidirectional. Therefore, layers of various sounds can be located and triggered more autonomously (Figures 4c). Whereas in *Stray*, the over-the-shoulder viewpoint places the camera behind and slightly detached from the character’s head position, featuring dynamic camerawork that adjusts according to movements by illustrating directional behavior. In this scenario, sound triggers are activated either by the actions of the protagonist or by camera movements and rotations, as determined by the designer’s choices (Figures 4a,b).

Through delineating zones, sound designers incorporate sound slots, adjust volume levels, apply fade-ins or fade-outs, modify rhythms and reverberations, and establish distinct tunes and themes for audibles in different open and enclosed spaces. Here, it is also worth noting that while “the visual side of gaming is a rather framed experience,” sonic experiences in audiovisual media are “borderless.”⁴⁸ Where the viewpoint

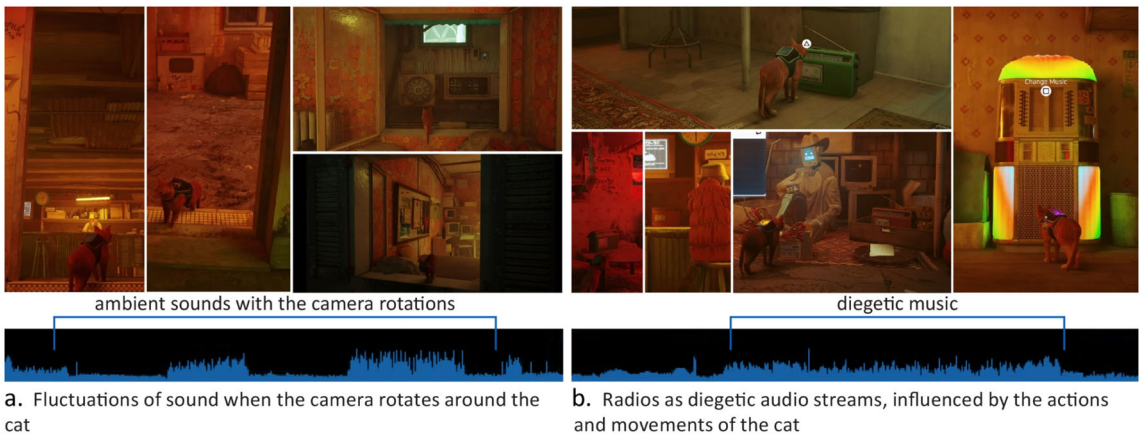


Figure 3
Stray, visualization of the soundboard, by the author, credit: BlueTwelve Studio.

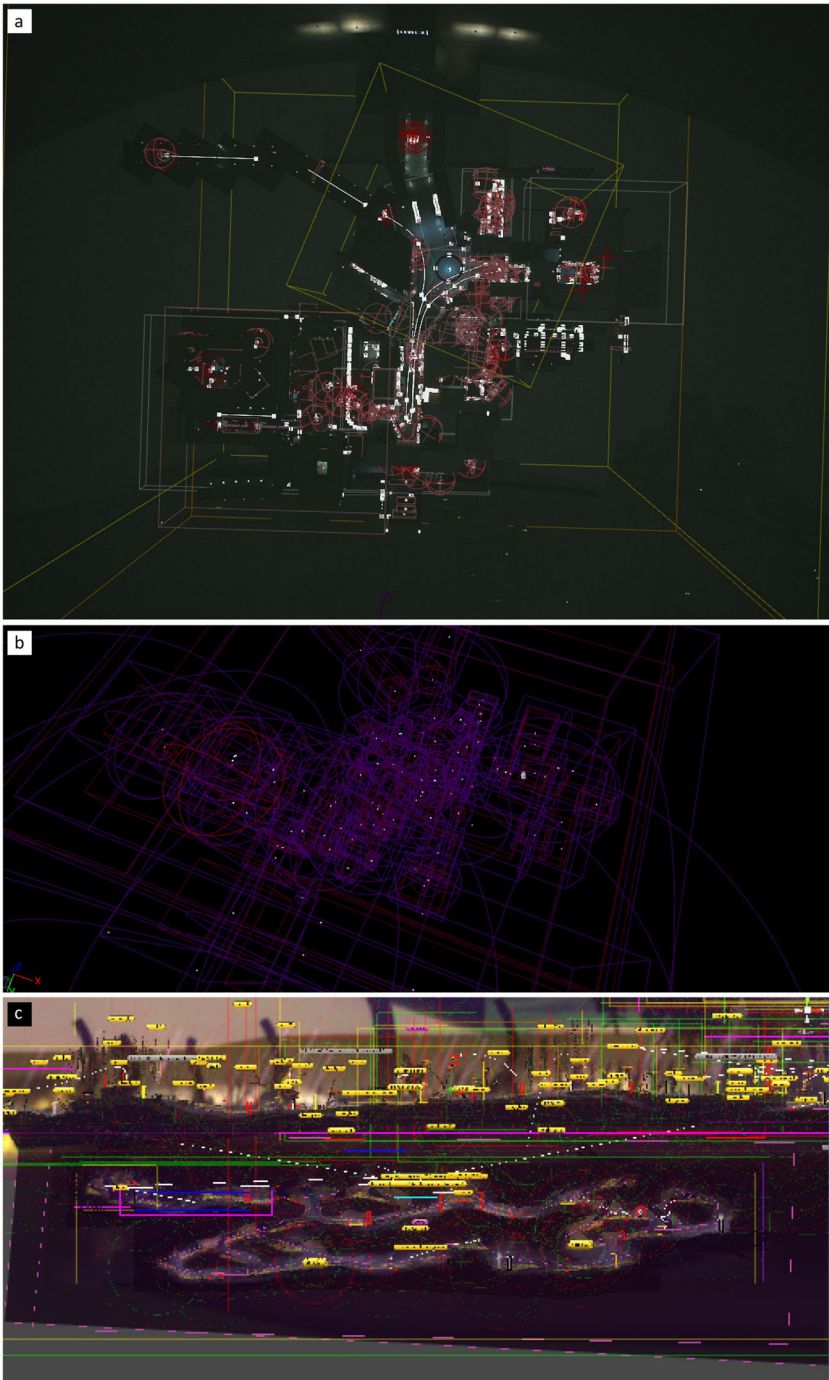


Figure 4
Comparison between the visual representation of a scene and its corresponding audio zone in a three-dimensional formatted game (a&b) to a visual map of audio sources in a two-dimensional game in which each yellow blob (short spatialized one-shot snippet) is an autonomous audio trigger (c), reproduced by permission: Van der Cruyssen.

is limited due to the camerawork and audio is not associated with how we ordinarily experience space, sound design acts as an informative source for further actions of the player and orienting them within a realm of defined space. Consequently, the combined impact of camerawork and sound zoning is crucial for portraying virtual spaces effectively in the experience.

The music in this game actively changes in response to the interactions of players with spaces and objects, and it is not guaranteed that any two players would hear the same musical sequences. This adds an element of unpredictability to the auditory experience and non-linear exploration within the game. Working as **a leitmotif, these aural themes serve to reinforce the idea that the music is about the game universe and its mysterious past.** Van der Cruyssen asserts that he “tried to really separate the gameplay from the music and have the music being an aesthetic element more than being very closely following the actions of the cat.”⁴⁹ Crafting a sonic connection between different scenes or locations helps to maintain a sense of continuity and emotional resonance during spatial explorations.

In *Stray*, synthesized sounds and music are closely integrated with visual elements to situate the player within a dystopian, futuristic world represented in three dimensions. This alignment not only reinforces spatial awareness but also fosters immersive listening by establishing a strong point-of-audition connection between the player and the on-screen character, evoking both spatial and subjective perception.

Journey: the music of the hero with a thousand faces (footage: <https://youtu.be/uYtRyOx6aM0>)

Inspired by Joseph Campbell’s concept of the monomyth, *Journey* undertakes a solitary quest through a vast and fantastical desert environment.⁵⁰ Designed in alignment with the narrative arc of The Hero’s Journey by leaving an ordinary life and setting forth on an adventure, this game outlines the experience of loneliness for the player’s character which “transforms into a cathartic experience through multiplayer.”⁵¹ Embarking upon a solo adventure with the opportunity to share it with an online stranger along the way, *Journey* narrates a wordless story of exploration, companionship, and adventure which designers aim to mirror through their creation of an immersive world and evocative music.

The music of the game concerning its virtual space is essentially atmospheric; it features sensory experiences by effectively conveying spatial qualities that immerse the player within its virtual world. The German philosopher, Gernot Böhm, presents a compelling perspective that music, particularly when combined with soundscapes, plays a central role in creating atmosphere within a space, triggering emotions—a phenomenon beautifully exemplified in *Journey*.⁵² Elsewhere, architectural theorist Michael Tawa identifies the “presentational” and “poietic” role of auditory experiences, and argues that sounds

fundamentally “carry or convey presence, to directly and concretely manifest realities, existences, and beings.”⁵³ Here, by exploring the interplay between the music and the space in *Journey* and drawing upon insights from phenomenological scholars emphasizing the significance of sound and music, we identify the bodily sense of self as the dominant experience of place; not just in physical environments but in representations of them, where audio as atmospheric elements contribute to its enaction.

Unlike most media forms, atmospheric soundtracks in interactive virtual environments of games “are objects with defined attributes, which have the power to play with the player’s moods, and must, in turn, react to the (subjective) input of the course of the game.”⁵⁴ While Nitsche argues that sound and soundscape “tell” the game environment to the player, the combination of world-building and music design in *Journey* communicates atmospheric characteristics of space.⁵⁵ It blurs the boundary between gameplay and emotional experience, offering a dynamic backdrop that responds to the player’s journey through the game.

The captivating atmospheric experience in *Journey* owes much to Austin Wintory’s careful musical direction. With a profound passion for both gaming and composing music, Wintory found his niche as a freelance music collaborator. Composing music for *Journey* was intertwined with the design process from its inception. Despite not having seen any initial concept art, Wintory began to formulate ideas on how to engage the player primarily through emotional evocation, with gameplay taking a secondary role. After their first conversation with Jenova Chen, the creative director, Wintory hummed melodies and hastily recorded them on his phone, which ultimately evolved into the iconic theme music of the game. He recalls that Chen once explicitly told him that “they were making changes to make the game [world] feel more like the music.”⁵⁶ The sound of music indeed became the central element that influenced the design of the game and its virtual environment.

The world-building of the game follows the same path to create an atmospheric experience. The architecture of the game portrays the esthetics of ruins, echoing the decayed structures of Greek and Roman civilizations, where rulesets dictated the design of buildings according to proportional systems. These remnants construct the architectural language of the game and their harmonious integration with the environment imbue the civilization of past eras. While the mountain serves as the ultimate destination, other landmarks assume the role of points of interest and guide the player through the open-world journey. Throughout the game’s quest, the character encounters architectural features that shape cascading sand in the desert, openings that allow light rays to penetrate the space, connecting bridges and thresholds that lead to the next path. Thus, these interventions define the architecture of this game not only as isolated buildings emerging at various phases of

The Hero's Journey but also as means for the designer to impact and shape the virtual environment in an atmospheric and meaningful way for the player to interact with.

Visualizations in this game are minimalist and poetic in nature. This game is essentially about the character's experience embodied through feelings. Therefore, to enhance emotional engagement while maintaining a streamlined visual approach, the integration of sound and music was necessary. This approach allowed for the communication of both gameplay actions to draw out potent feelings at key points in the narrative and atmosphere, facilitating a deeper connection between the player and the virtual world.

Wintory describes *Journey* as a silent film, "without dialogue and heavy-handed sound effects like big monsters and things to sort of distract you."⁵⁷ Within the game, music serves as the primary mechanism for storytelling in the absence of dialogue or text, imparting the tone of each scene to the player. Game music must react to players' actions and in-game space design; if this interactive component fails to fulfill its role, it runs the risk of becoming mere musical wallpaper and "can even be turned off with no to little effect on the gameplay."⁵⁸ Therefore, the challenge for Wintory was preventing the music from becoming wallpaper, whilst conversely not allowing it to distract from the gameplay. Nonetheless, music in this game relates to the context in such a way as to refer beyond the sole utilitarian aspects of performing actions: to convey the symbolic and constitutive character of spaces to the player-listener.

What Wintory means is that despite the gameplay occurring in a desert, the music deliberately avoids drawing on orientalist imagery associated with culturally desert-adjacent music. He explains "this was a source of inspiration I intentionally rejected ... sort of like, exotifying something in a way that's superficial and uninteresting."⁵⁹ He emphasizes that the desert serves as a backdrop for its emotional implication, symbolizing a place intrinsically hostile to most forms of life, thus evoking a profound sense of isolation. With the foundational concept of the narrative structured around the various stages of the Hero's Journey, it became evident that the theme music needed to "be very simple that could bend and change shape, and would be obviously recognizable in other contexts."⁶⁰ The theme music, which later became known as *Nascence*, acts as a subtle yet powerful reminder of the player's enduring a journey. For example, during the time that the player ventures into the scary and hostile environment of the mountain, the theme music emerges, serving as a poignant reminder that the player and character are enduring this journey together.

Due to the budget constraints, the team could only record approximately thirty minutes of music with the orchestral strings which constituted about half the score at most. Therefore, the team sought a strategic approach to this limitation: "what if the orchestral strings are a measurement of how far you are from the mountain," a central motive to

the game's narrative arc.⁶¹ At the outset, as players embark on their journey in the vast, yellow-hued expanse of the first stage, The Graveyard, the absence of strings signifies the maximum distance from the mountain, evoking a sense of solitude (Figure 5a). With no strings playing and the gradual addition of string instruments in subsequent areas, the non-diegetic music acts as a geo-locator, mirroring the game's staged design.

Each instrument symbolizes a character in the game. During the initial stage, where the player traverses alone, the solo cello resonates to portray the protagonist. Upon encountering a companion, the auditory landscape expands to include the flute to symphonically represent the companion. In the same way, the harp and viola embody ancestral figures and the Serpent (18th century horn) corresponds to enemies (Figure 5b). The first encounter with the strings occurs in the third area, The Desert, symbolizing progression closer to the mountain's domain. This deliberate placement aligns with the game's thematic elements, with each area embodying distinct challenges and atmosphere. Wintory plays with the pace and rhythm of the theme music in different places or fractures it, yet the listener can still discern the iteration of its presence and make a connection with the experience of the character in the game.

In The Desert, the main gameplay action involves a thrilling voyage through surfing on the sand. This dynamic gameplay experience is not only visually captivating but also deeply immersive due to its synergy with audio elements. The music pulsates with energy, perfectly synchronized with the action, enhancing the emotional journey and fully immersing the player in the exhilarating experience (Figure 5c). At one key moment, the music, environmental design, and the gameplay converge intuitively to become one concise conversation. Playing around with high-energy music within the established musical vocabulary of *Journey*, Wintory developed a rhythm that elegantly connects with the environment, being fast and exciting, yet navigating away from tension or stress building that typically signifies danger or a sense of battle. To align this energetic piece with the scenery, the so-called Parallax Tunnel swings the camera to a locked side-view, and the player's progress is rewarded with a gorgeous vista of the mountain framed by the setting sun. Each element contributes to the immersion of the game, culminating in a truly impactful moment that leaves the player with a resounding and unforgettable impression of the experience they have just been a part of (Figures 4d).

The final and most memorable sequence of the game is supported by an orchestral score, appropriately titled "Apotheosis." This piece builds on the player character's motif of airy and flowing movement, exploding from a sullen depth into a racing surge of energy and liveliness, reflecting the climb from a frozen tundra through to the mountain objective, positively bristling with life. This climactic moment has solidified the music of *Journey* as quintessentially orchestral to its wider

audience. However, to Wintory, this is a “fallacy of composition” as he also emphasizes that he employed various other techniques throughout his compositions (Figures 4e).

Within the game, the soundscape and sound effects respond directly to environmental changes and actions, nevertheless, the creative direction of the music is non-diegetic. The music plays a different role

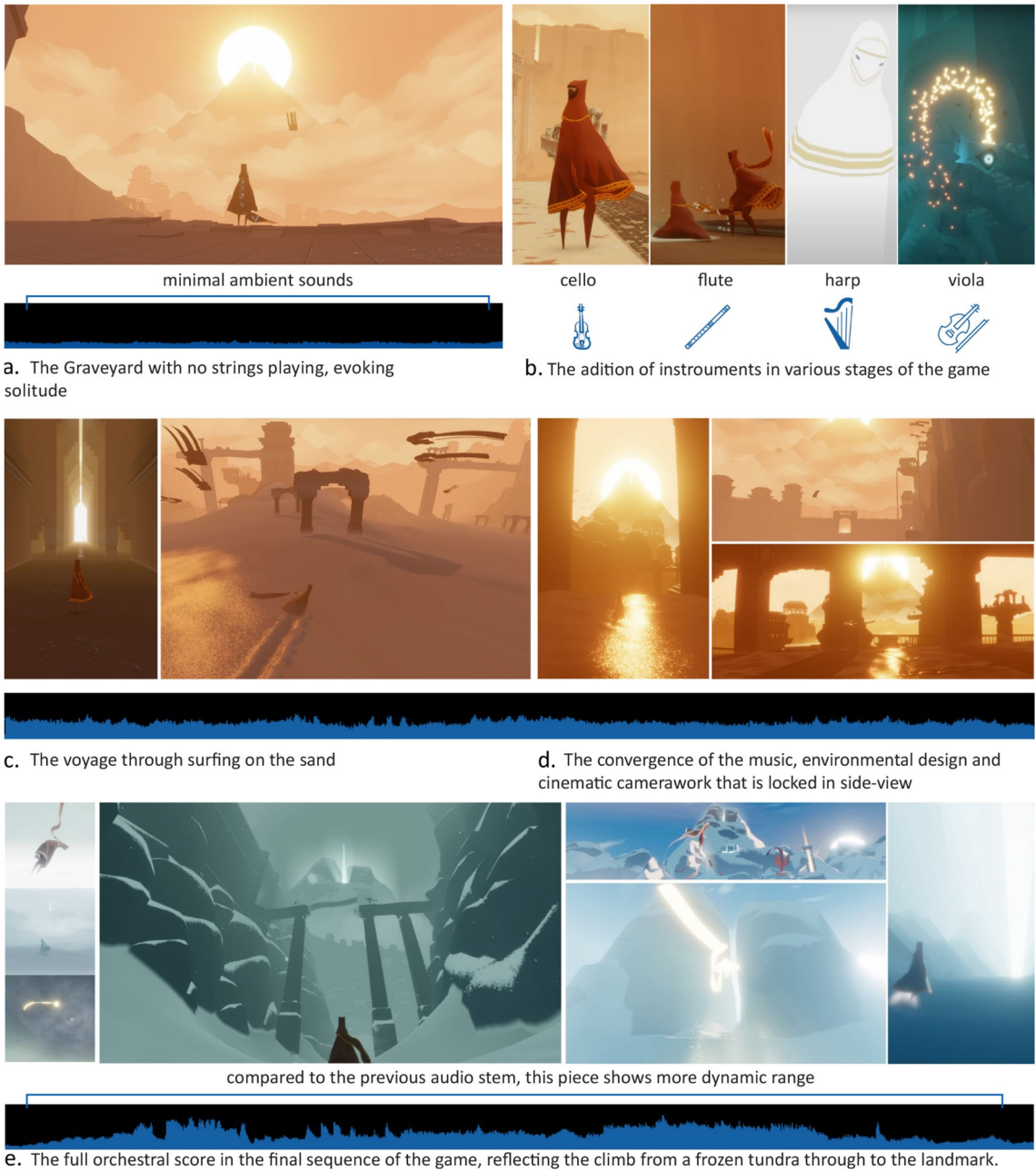


Figure 5
Journey, visualization of the soundboard with notes, by the author; soundboard: <https://youtu.be/UUbvQW3aJpk>, credit: thatgamecompany.

functionally every time to embody the essence of *Journey*'s design ethos. Wintory explains that:

Sometimes the goal is to lean into the environment, but sometimes it's about something different ... honestly, I would say capturing the environment really wasn't a priority for *Journey*. It was more like capturing what the environment evokes, which is not the same thing. It's more ... the environment creates an emotion. So, then I want to grab that emotion. It's not about grabbing the environment directly.⁶²

In *Journey*, the music primarily supports narrative understanding and is "adaptive to player movement over its world."⁶³ Although the music is not directly tied to specific in-world sources, it forms powerful new connections with the environment through a phenomenon known as synchresis defined as "the spontaneous and irresistible mental fusion, completely free of any logic, that happens between a sound and a visual when these occur at exactly the same time."⁶⁴ This fusion allows sound to acquire new meanings through its association with visuals, even if those sounds originate from unrelated or "schizophonic" sources.⁶⁵ In *Journey*, the music, though not stereotypically desert-like, gains new significance by linking to its visuals, creating a unique embodied audiovisual experience.

Conclusion

Auditory elements of sound, soundscape, and music animate and narrate game worlds, complimenting the visuals of environmental storytelling. They contribute to virtual world-building to situate players in a digital space, helping to understand and map their surroundings intuitively to aid immersion; audio reveals the game space to the player. Game audio designers create, curate, and integrate contextual sound and music, connecting visuals and audio to actions of the player, demonstrating their impact on the world through its responsiveness. Splicing sound, soundscape, and music into the design of the world makes the audio become an innate part of the environment, simultaneously enhancing spatial awareness and adding richness to the immersive, digital realm.

Three indie-developed videogames, *Rain World*, *Stray*, and *Journey*, narrate and navigate their wordless stories through environmental visions, actions, and audio. Sound and music assume an intimate role in crafting their world-building. They portray alternative worlds with unprecedented settings where audio contributes to the cultivation of another sense of place within the gaming environment. Creating an alternative experience of reality, and not an alternative reality alone, is the cornerstone of many videogames using narrative structures to design their worlds. Their interactive images foster immersion by providing an

element of familiarity in audio, grounding players' understanding of speculative or unfathomable virtual environments. As such, sound design in games can be seen as a fundamental atmospheric principle of interactive spaces affecting experience, exploration, manipulation, and perceptual representation of space.

Each of the case studies in this article exemplifies the reciprocal relationship between audio and visual representations, each highlighting a distinct quality. In *Rain World*, a crucial element of sound design and music is the synergy with visuals to evoke a strong sense of immersion through imagination in the virtual environment. In the three-dimensional space of *Stray*, integrating and defining volume, rhythm, pace, beginning, and end of audio are essential for crafting the game's atmosphere. It involves creatively designing soundscapes that complement and enhance the visuals, utilizing sound curation techniques to achieve a deeper immersion. Symbolic and metaphoric use of music in *Journey* clearly elucidates how atmospheric audio is linked to the notion of immersion in virtual environments. Intentionally curated and integrated sounds are used as evocative elements to express immersion and guide the comprehension of a represented space.

These examples are different in visual styles and distinct in the purposes that their audio serves. However, each case study still uses sound effects, soundscape, and music as design tools to immerse and engage the player, systematically connecting to emotional stimuli to influence player presence. Digitally created sound effects and musical traditions in a videogame milieu align with space, time, and culture, providing a deep reception of their creations—feeling our subjective and personal presence in their environments.

This article begins to address the connection between sound and virtual space, exploring the role audio plays in creating spatial immersion—a key tactic for architects and designers who intend to design virtual realms. Sound, soundscape, and music help position the player within a specific time and space. Coupled with the camerawork, they establish a distinct point-of-audition that not only situates the user differently within a space, but also drives the narrative, shapes the atmosphere, and influences the perception of a digital world. As seen in *Rain World* and *Stray*, diegetic audio reveals spatial qualities and communicates the relationship between the player character and the environment. These audio cues contribute to create mundane or fantastical settings that feel authentic, with sounds and music believably emerging from the world itself. In contrast, *Journey* utilizes non-diegetic music to craft an abstract emotional connection to its environment. This method allows audio to act as a narrative and spatial guide, shaping new meanings and emotional resonances without becoming wallpaper music. These examples demonstrate how audio in relation to the environment can add layers of meaning to positioning, storytelling, and perception. By mediating how players experience and interpret virtual spaces, sound

encourages exploration, influences movement, and invites moments of pause to engage with the environment—ultimately deepening immersion and engagement with the virtual world.

Gameography

Journey (Thatgamecompany, 2012).

Rain World (Videocult, 2018).

Stray (BlueTwelve Studio, 2022).

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Data Availability Statement

The data that support the findings of this study from the interviews are available on request from the corresponding author, Dorsa Kafili. The data are not publicly available due to the Ethics policy of the University of Melbourne, where this research took place. Name of the ethics committee: Human Ethics Team of the University of Melbourne; HumanEthics-Enquiries@unimelb.edu.au. Approved number: 2023-25519-37534-3

Supplemental Material

Videogame footage and original gameplay are uploaded to YouTube, with links provided in the article for accessing the relevant information.

Notes

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 13. David Byrne, *How Music Works* (San Francisco: McSweeney's Publishing, 2012).
 14. Blesser and Salter, *Spaces speak, are you Listening?*
 15. *Ibid.*, 5.
 16. James Therrien (narrative, world-building, music, and sound designer), in discussion with the authors, June 2023.
 17. Karen Collins, *Playing with Sound: A Theory of Interacting with Sound and Music in Video Games* (MIT Press, 2013), 3.
 18. *Ibid.*
 19. Rather than being a continuous linear world exploration, *Metroidvania* levels in game design are applied to videogames where exploration unfolds in various directions, occasionally necessitating backtracking to progress.
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 27. In indie games, design and production teams are typically small, with job titles being more flexible. At a minimum, game teams usually include a level designer, a sound designer, and a programmer.
 28. Therrien, interview. Here, the terms "level" and "level design" refer to the environment, designers use environmental design and level design interchangeably.
 29. Chiptune music is a form of videogame vintage music that represents the nostalgic soundtracks of early videogames, crafted primarily through repurposed classic videogame hardware. It serves as both a performance space for electronic music and a medium for recreating the retro electronic soundscapes that defined the era.
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 47. Chion, *Audio-Vision*, 1994.
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 49. Van der Cruyssen, interview.
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